Unit: Unit 6: Systems of Linear Equations and Inequalities
Topic: Linear Inequalities with Two Variables

Name: ________________
Date: ____________

Quiz (Carolina Reaper)

Q1. What is the graph of the inequality $x - 2y > 3$?

(A) Solid line with y-intercept at $(0, -1.5)$
(B) Dashed line with y-intercept at $(0, -1.5)$
(C) Solid line with y-intercept at $(0, 1.5)$
(D) Dashed line with y-intercept at $(0, 1.5)$
(E) No boundary line

Q2. A transportation company has a fleet of trucks that can carry a maximum of $2x + 5y$ tons, where x is the number of vehicles and y is the number of drivers. If each truck can carry at least 4 tons, what inequality represents this situation?

(A) $2x + 5y \geq 4$
(B) $2x + 5y \leq 4$
(C) $2x - 5y \geq 4$
(D) $2x - 5y \leq 4$
(E) $x + y \geq 4$

Q3. The solution region of the inequality $y < 2x - 1$ is bounded by a

(A) Solid line
(B) Dashed line
(C) Horizontal line
(D) Vertical line
(E) No boundary line

Q4. A logistics company needs to deliver packages to 3 different locations. The time it takes to travel from location 1 to location 2 is x hours, and from location 2 to location 3 is y hours. If the total delivery time is at most 5 hours, what inequality represents this situation?

(A) $x + y \leq 5$
(B) $x + y \geq 5$
(C) $x - y \leq 5$
(D) $x - y \geq 5$
(E) $xy \leq 5$

Q5. Which of the following inequalities has a solution region that is unbounded in the first quadrant?

(A) $x + y > 2$
(B) $x + y < 2$
(C) $x - y > 2$
(D) $x - y < 2$
(E) $xy > 2$

Q6. The inequality $y \leq -2x + 3$ represents a

(A) Solid line with y-intercept at $(0, 3)$
(B) Dashed line with y-intercept at $(0, 3)$
(C) Solid line with y-intercept at $(0, -3)$
(D) Dashed line with y-intercept at $(0, -3)$
(E) No boundary line

Q7. A shipping company has a truck that can carry a maximum of 1500 pounds. The truck is currently carrying x pounds of boxes and y pounds of furniture. If the weight of the boxes is at least 200 pounds more than the weight of the furniture, what inequality represents this situation?

(A) $x \geq y + 200$
(B) $x \leq y + 200$
(C) $x + y \leq 1500$
(D) $x + y \geq 1500$
(E) $x - y \geq 200$

Q8. The solution region of the inequality $y > x^2 - 4$ is bounded by a

(A) Solid parabola
(B) Dashed parabola
(C) Solid line
(D) Dashed line
(E) No boundary curve

Open Ended Questions (FRQ)

Q9. Graph the inequality $y < 2x - 1$ and shade the solution region. Label the x- and y-axes and provide a title for the graph.

Q10. A traffic flow model is given by the inequality $x + 2y \leq 10$, where x is the number of cars and y is the number of trucks. If there are currently 3 cars and 2 trucks on the road, does this situation satisfy the traffic flow model? Provide a justification for your answer and graph the inequality.

Chef's Tips

- Remind students to label the axes and provide a title for the graph.
- Encourage students to use a ruler to draw the boundary lines.
- Emphasize the importance of shading the correct region.